Multi-wavelength Astronomy

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

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1. One spectrum, many messengers

Astronomy is mostly the art of catching light. Radio waves, infrared, visible light, ultraviolet, X-rays and gamma rays are all electromagnetic radiation; they differ in wavelength, frequency and photon energy. The relations are

$$c = \lambda\nu, \qquad E = h\nu = \frac{hc}{\lambda}. \tag{1}$$

Use $c = 2.998 \times 10^{8}\ \mathrm{m\,s^{-1}}$, $h = 6.626 \times 10^{-34}\ \mathrm{J\,s}$, and $hc = 1240\ \mathrm{eV\,nm}$. A convenient rule is that shorter wavelength means higher photon energy. A 500 nm photon carries about 2.48 eV, while a 1 keV X-ray photon carries about 400 times more energy.

Different bands are not merely "colours." They trace different physical conditions: cold dust emits in the far-infrared, stars peak in the optical or ultraviolet, hot gas emits X-rays, and relativistic particles can produce radio synchrotron emission.

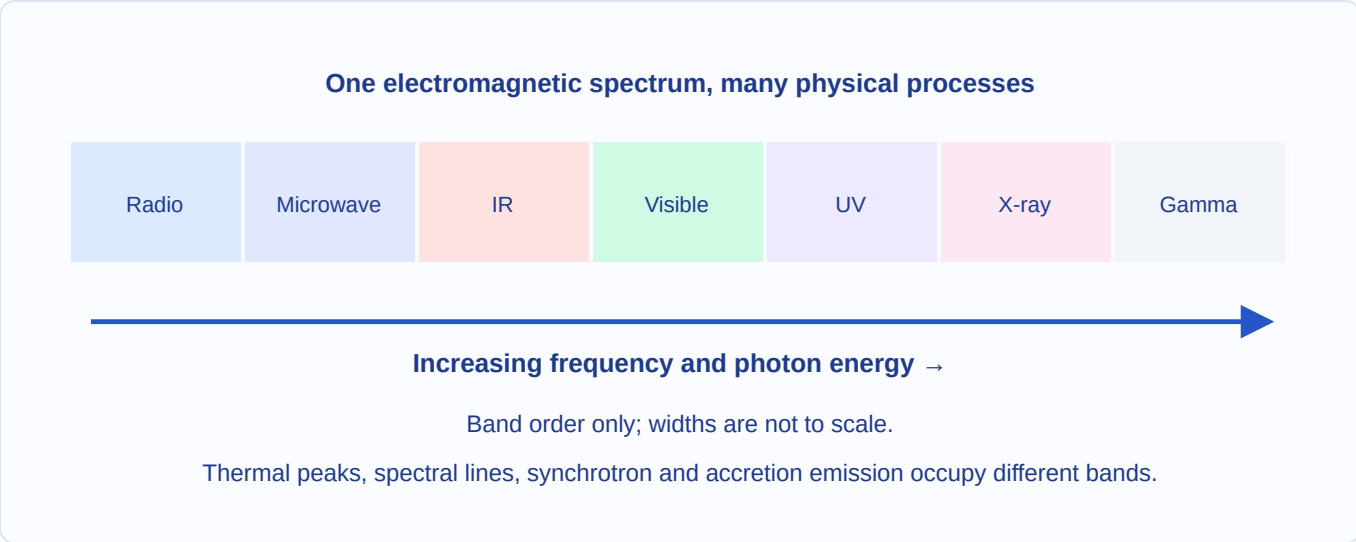


Figure 1. Band boundaries are conventional. A complete picture of an object usually requires several wavelength regimes.

2. Thermal radiation and Wien's law

A dense warm object emits a broad thermal spectrum. For an ideal blackbody, the wavelength of maximum emission is approximately

$$\lambda_{\max} T = 2.898 \times 10^{-3}\ \mathrm{m\,K}. \tag{2}$$

The Sun, near 5800 K, peaks around 500 nm. A 300 K planet peaks near 10 μm. A 10 K dust cloud peaks near 290 μm. The peak is only a guide: real objects have absorption lines, emission lines, dust features and non-thermal components.

The total thermal power per unit area scales as T^4. Doubling the temperature increases the emitted flux by a factor of 16. This is why temperature is such a powerful diagnostic.

3. Redshift moves the whole spectrum

Cosmological redshift stretches every wavelength by the same factor:

$$\lambda_{\mathrm{obs}} = \lambda_{\mathrm{rest}}(1 + z). \tag{3}$$

A rest-frame ultraviolet line at 121.6 nm observed at 608 nm implies $z = 4$. The same factor applies to a thermal peak. This is why high-redshift galaxies are often studied in the infrared: their rest-frame optical and ultraviolet light has been shifted out of the visible band.

Redshift does not change the identity of a spectral line; it changes the observed wavelength. Identifying several lines with the same z is much stronger evidence than matching one feature.

4. The atmosphere is a selective filter

The ground is not equally transparent at all wavelengths. The optical window is broad, radio windows are excellent, and some near-infrared windows are usable from dry sites. Much ultraviolet, X-ray and gamma-ray astronomy must be done from space. Water vapour, ozone, oxygen and carbon dioxide shape the transmission.

This is not a minor engineering detail. It determines which questions can be answered from the ground and which require balloons, rockets or satellites. A “missing” band is often an atmospheric effect, not an absence of astrophysical emission.

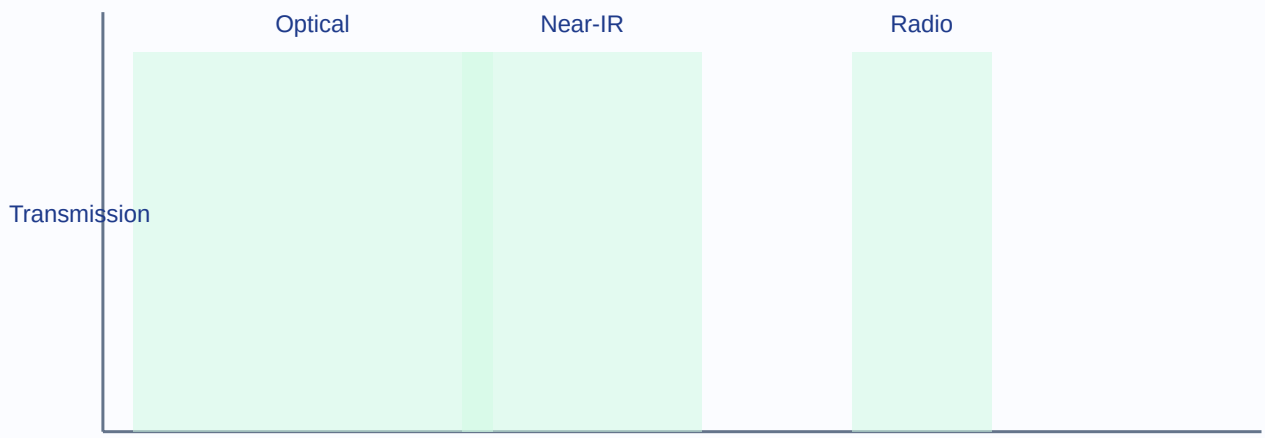


Figure 2. Optical and radio windows are broad; much UV, X-ray and gamma-ray astronomy requires space telescopes.

5. Extinction, emission and diagnostic power

Dust scatters and absorbs short wavelengths more efficiently than long wavelengths. In magnitudes, an extinction $A_V = 2$ reduces the observed flux by $10^{0.4A_V} = 6.31$. Infrared observations can therefore see through dust that blocks visible light.

Spectral lines add chemical and kinematic information. The Doppler shift gives velocity; line ratios can give temperature, density or ionization state. Radio 21 cm emission traces neutral hydrogen; molecular lines trace cold gas; X-ray lines trace very hot plasma.

6. Build a multi-wavelength argument

A good multi-wavelength study asks what each band contributes. Optical images show stars and morphology; infrared reveals dust and cool objects; radio traces gas and magnetic fields; X-rays reveal hot gas and compact objects. The bands should be compared with matched angular resolution, calibration and aperture.

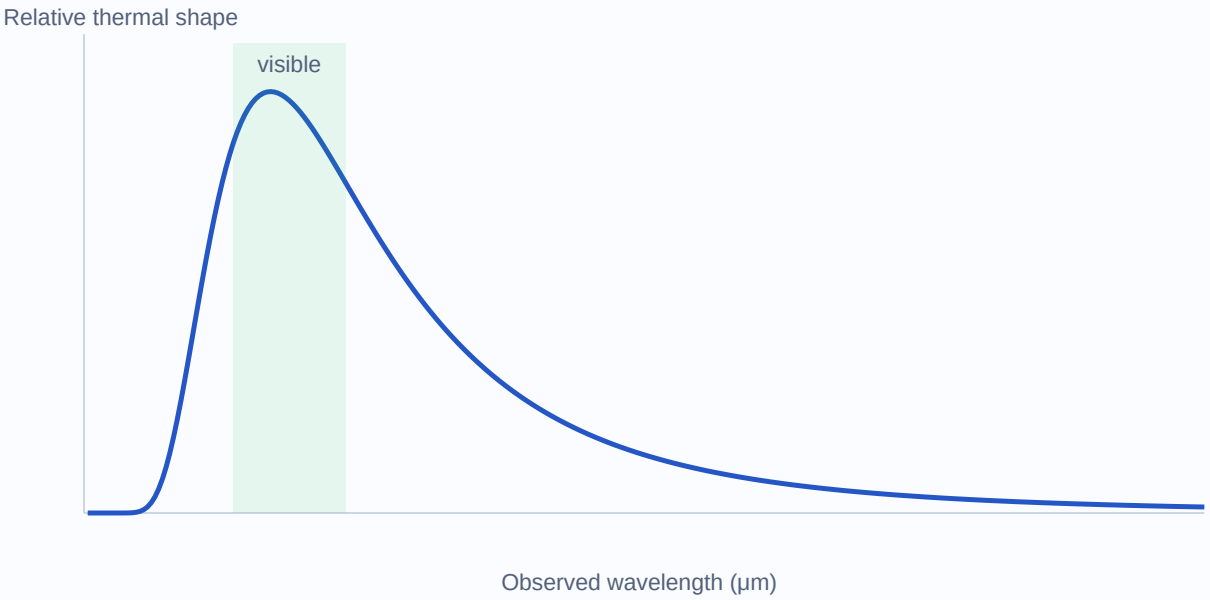
INTERACTIVE EXPERIMENT

Where does the light emerge, and can the ground see it?

A thermal source is shifted by redshift. The peak is only a guide: real sources also emit non-thermal radiation, and atmospheric windows are wavelength dependent.

Rest-frame peak	Observed peak	Band
0.500 µm	**0.500 µm**	**Visible**

Photon energy at peak	Ground access
2.482 eV	**Good optical window**



The peak is a guide, not the whole spectrum. Redshift moves every wavelength by the same factor (1+z); atmospheric access depends on the observed wavelength.

Try this. Increase z until an optical peak moves into the infrared. Why does this help high-redshift astronomy?

Snapshot of the experiment. Open the HTML edition to change the controls.

A reliable method. Convert everything to the same units. Ask whether the quoted wavelength is rest-frame or observed. Check whether the atmosphere transmits it. Then ask which physical process can produce photons of that energy.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — A visible photon. Find the frequency and energy in joules and electronvolts of a 500 nm photon.

Worked solution →

[1] Problem 2 — Two thermal peaks. Use Wien's law to estimate the temperatures of blackbodies peaking at 500 nm and 10 μm.

Worked solution →

[2] Problem 3 — A shifted line. Lyman-α has rest wavelength 121.6 nm. It is observed at 608 nm. Find the redshift and the observed wavelength of a rest-frame 500 nm feature from the same galaxy.

Worked solution →

[1] Problem 4 — Magnitude contrast. Two sources differ by 5 magnitudes in the same band. What is their flux ratio? Which is brighter?

Worked solution →

[1] Problem 5 — Ground or space? Classify observations at 21 cm, 550 nm, 200 nm and 1 keV as generally ground-accessible or space-required. Give one physical reason for each.

Worked solution →

[2] Problem 6 — Dust penalty. A star suffers $A_V = 2$ mag of visual extinction. By what factor is its visual flux reduced? Why might an infrared observation be preferred?

Worked solution →

[2] Problem 7 — Counting X-rays. An X-ray detector receives 1.0×10^{-15} W in 1 keV photons. Estimate the photon arrival rate. Why is X-ray astronomy often photon-counting rather than intensity-measuring?

Worked solution →

[2] Problem 8 — Radio versus optical resolution. Estimate the Rayleigh limit of a 25 m radio dish at 21 cm and a 10 cm optical telescope at 500 nm. Express both in arcseconds.

Worked solution →

[3] Problem 9 — A high-redshift thermal peak. A galaxy at $z = 2$ contains stars approximated by a 10,000 K blackbody. Find the rest-frame and observed peak wavelengths. In which broad band is the observed peak?

Worked solution →

[3] Problem 10 — Design a campaign. A dusty star-forming galaxy is faint in the optical but bright in the infrared and radio. Propose three wavelength regimes and state what each would constrain. Mention one resolution or calibration issue.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.